

MANIKANDAN B

Aminjikarai, Chennai, Tamil Nadu | +91 7358295991 | manikandan270706@gmail.com
linkedin.com/in/manikandan-b-aa45032a0 | github.com/Manikandan-027

PROFESSIONAL SUMMARY

Final-year B.Tech AI & Data Science student (CGPA 8.6) building production-style, full-stack AI systems across NLP, Computer Vision, IoT, and Retrieval-Augmented Generation (RAG). Hands-on with LLM pipelines (LangChain, FAISS, Ollama), Python, and full-stack development (FastAPI, React, Next.js, Node.js, Flask). Seeking AI Engineer, ML Engineer, or Full-Stack Developer roles to build reliable, scalable AI-driven products.

EDUCATION

B.Tech, Artificial Intelligence & Data Science: Velammal Engineering College, Anna University, Chennai | CGPA: 8.6/10.0 | 2023 – 2027

Higher Secondary Certificate (HSC): Thiru Vi Ka Higher Secondary School, Chennai | 87%

Secondary School Leaving Certificate (SSLC): Thiru Vi Ka Higher Secondary School, Chennai | 100%

KEY SKILLS

Languages: Python, SQL, C (basics)

LLM / GenAI: LangChain, RAG Pipelines, FAISS, Vector Databases, Ollama, Prompt Engineering, Hugging Face Transformers (DistilBERT)

ML & Data Science: Scikit-learn, Pandas, NumPy, TF-IDF, NLTK, Matplotlib, Seaborn

Computer Vision: OpenCV

IoT & Hardware: ESP32, MPU6050, Neo-6M GPS Module, Sensor Fusion, Cloud Database

Frameworks & Deployment: Flask, FastAPI, React, Next.js, Node.js, Express.js, React Native, Streamlit, Git, JWT Authentication

Tools: Jupyter Notebook, Google Colab, CustomTkinter

PROJECTS

Enterprise AI Knowledge Assistant | *Python, FastAPI, React, LLM (Ollama), Vector Database, JWT Auth*

- Built a full-stack AI knowledge platform where users upload PDF/DOCX/TXT documents, which are chunked, embedded, and stored in a vector database for semantic retrieval.
- Built an AI chat interface that retrieves relevant document context and generates answers via an LLM through Ollama, with source references, document management, analytics, system health monitoring, and backup/restore.
- Implemented secure JWT-based authentication and user login/registration; backend in Python FastAPI, frontend in React.

InterviewArena – AI-Powered Interview Preparation Platform | *Python, FastAPI, T5 (Grammar Error Correction), Hugging Face Transformers, Next.js, React, REST APIs*

- Developed InterviewArena, a full-stack interview preparation platform delivering interactive coding/quiz practice alongside an AI-assisted speaking coach.
- Built a dedicated Python FastAPI ML microservice with a fine-tuned T5-based Grammar Error Correction (GEC) model that identifies grammatical errors, generates corrections, classifies weak areas, and delivers contextual feedback.
- Integrated the ML service with a Next.js/React frontend through REST APIs for seamless, real-time response analysis.
- Designed validation and safety logic to filter out unreliable grammar corrections, ensuring only high-confidence feedback reaches users.

TruthLens AI – Fake News Detection Platform | *Python, DistilBERT, Transformers, Explainable AI, Flask*

- Built an AI-powered fake news detection platform using a fine-tuned DistilBERT model, achieving 97.95% accuracy and 97.98% F1-score on 1,267 test articles.
- Added real-time fact-checking, explainable-AI output, analytics, user authentication, and automated report generation behind a modern responsive UI.

IoT-Based Smart Pothole Detection & Warning System | *ESP32, MPU6050, Neo-6M GPS, Machine Learning, Cloud Database*

- Built a pothole detection system combining an ESP32 + MPU6050 vibration sensor for shock detection with a phone-mounted camera feeding an ML model that classifies pothole severity (normal, small, dangerous).
- On confirming a dangerous pothole, captured its location via a Neo-6M GPS module and uploaded severity + coordinates to a cloud database, surfaced on a real-time dashboard.
- Delivered early warnings to nearby vehicles through the dashboard, voice alerts, and optional LED/buzzer notifications to reduce accidents and aid road maintenance.

INTERNSHIP EXPERIENCE

Developer Intern | Virtual Tech Services, Ambattur | 1 Month

- Built the Fake News Detection application end-to-end: data preprocessing, model training, evaluation, and deployment support.
- Collaborated with the development team through the full build cycle, from data collection to final testing.

CERTIFICATIONS

- Artificial Intelligence Fundamentals – IBM SkillsBuild
- Big Data Processing with Apache Spark (PySpark) – IBM
- Natural Language Processing – NPTEL

LANGUAGES

English (Fluent) • Tamil (Fluent)